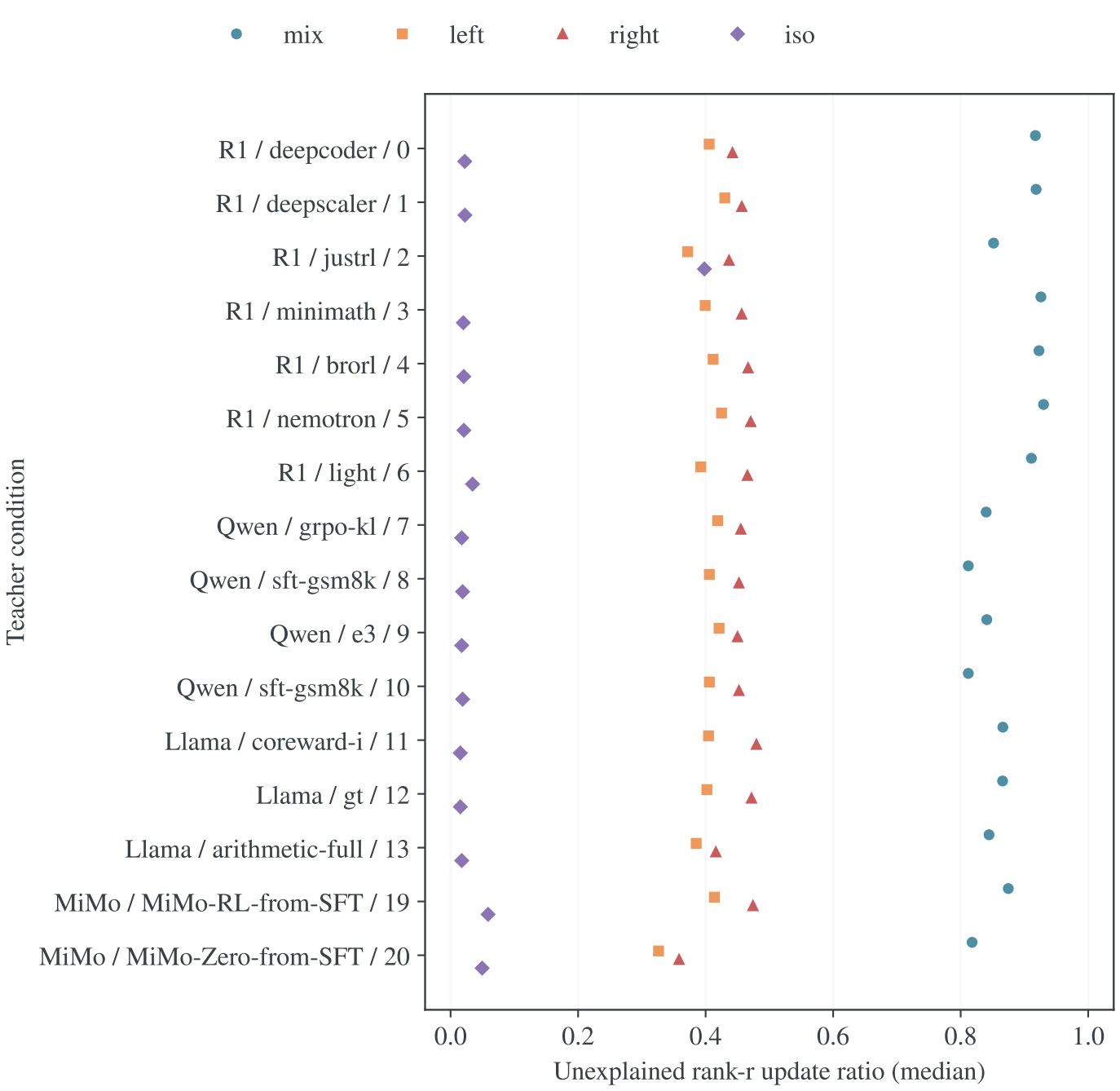

r = floor(0.9q); ratios use Frobenius norm, not squared energy.